D3.3 Homeostasis

NAME	DATE	TOTAL
		_____ / 48

About 50 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **DEFINE** homeostasis. [D3.3.1] [1]

2 **LIST** three variables that are controlled by homeostasis in humans. [D3.3.1] [2]

3 **STATE** the type of cell that secretes insulin, and where these cells are found. [D3.3.3] [2]

4 **STATE** the part of the brain that acts as the temperature control centre. [D3.3.5] [1]

5 **STATE** the gland that secretes thyroxin. [D3.3.5] [1]

6 **DISTINGUISH** between glucagon and glycogen. [D3.3.3] [2]

Section B: Understanding

13 MARKS

7 **EXPLAIN** why homeostasis uses negative feedback rather than positive feedback. [D3.3.2] [2]

8 **EXPLAIN** how blood glucose returns to normal after a period of fasting. [D3.3.3] [4]

9 **COMPARE AND CONTRAST** type 1 and type 2 diabetes. [D3.3.4] [4]

TYPE 1	TYPE 2

10 **EXPLAIN** how sweating cools the body. [D3.3.6] [3]

Section C: Working with data

18 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D3.3.3, D3.3.4]

Two people fasted and then drank the same dose of glucose. Their blood glucose was measured.

TIME / HOURS	0	0.5	1	2	3	4
PERSON P / MG PER 100 CM³	82	140	150	100	85	80
PERSON Q / MG PER 100 CM³	145	230	290	330	290	240

a **CALCULATE** the percentage increase in glucose in person P from 0 to 1 hour. [2]

b **COMPARE** the responses of P and Q. [3]

c **EXPLAIN** the fall in blood glucose in person P after 1 hour. [3]

d Person Q's blood insulin was found to be above normal. **DEDUCE** the type of diabetes, giving a reason. [2]

12 [D3.3.5, D3.3.6]

A person moved from a warm room into a cold room for 40 minutes. Their skin temperature and core temperature were recorded.

TIME / MINUTES	0	10	20	30	40
SKIN TEMPERATURE / °C	33.5	30.0	28.0	27.0	26.5
CORE TEMPERATURE / °C	37.0	36.9	36.8	36.9	37.0

a CALCULATE the change in skin temperature over 40 minutes. [1]

b EXPLAIN how a fall in skin temperature benefits the body. [2]

c EXPLAIN why core temperature stays almost constant, with two other responses. [3]

d STATE the receptors that detect the fall in temperature of the skin. [1]

e SUGGEST why a newborn baby is at greater risk in the same room. [1]

Section D: Longer answers

8 MARKS

13 DESCRIBE the role of thyroxin in regulating body temperature, starting at the hypothalamus. [D3.3.5] [4]

14 OUTLINE four ways to reduce the risk of type 2 diabetes, or to manage it. [D3.3.4] [4]